Dr Aarzoo Dhiman

Research Scientist | Natural Language Processing and Machine Learning
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Profile

Natural Language Processing (NLP) and machine learning research scientist (PhD, IIT Roorkee; postdoc, University College London). Works on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), speech recognition, transformer text classification, event and misinformation detection in social media (up to 107.9M posts). Papers in npj Digital Medicine, Expert Systems with Applications, ACM TALLIP, IEEE Access, IEEE BigData.

Experience

Teaching Fellow Nov 2023 – Present
Centre of Excellence for Data Science, Artificial Intelligence and Modelling (DAIM), University of Hull Hull, UK

- Seconded as Co-Investigator (PI: Dr Will Jones) to a £41,707 Humber and North Yorkshire Cancer Alliance award (Sep 2025 – Mar 2026), building an on-device AI system for breast cancer MDT meetings with Hull University Teaching Hospitals NHS Foundation Trust; lead and corresponding author.
- First author of a predictive machine learning study of suicide-related risk on 538 anonymised NHS mental health records with ICD-10-coded diagnoses (scikit-learn, XGBoost, SHAP, mixed-effects models), with South West Yorkshire Partnership NHS Foundation Trust; under review at Archives of Suicide Research.

Postdoctoral Research Fellow Jun 2021 – Nov 2023
Department of Computer Science, University College London (UCL) London, UK

- Owned implementation, data collection, analysis and testing for a social media study of COVID-19 household transmission; first-author paper in npj Digital Medicine (2024).
- Worked with epidemiologists and Virus Watch; compared estimates with UK Health Security Agency data.

PhD Research Scholar Mar 2017 – May 2022
Indian Institute of Technology (IIT) Roorkee Uttarakhand, India

- Researched NLP for public health and government policy on Twitter (supervisor: Prof. Durga Toshniwal): 3 journal articles and 4 conference and workshop papers.
- Partnered with Intel India on misinformation detection; selected as Intel Student Ambassador (2020) and awarded Intel conference-fee funding (2021).

Selected Research

On-device AI for breast cancer multidisciplinary team (MDT) meetings, University of Hull with Hull University Teaching Hospitals NHS Foundation Trust 2025 – 2026

- Built and deployed an automatic speech recognition (ASR), LLM and RAG pipeline on one NVIDIA Jetson AGX Orin edge GPU (Whisper large-v3; quantised MedGemma 27B via llama.cpp, CUDA, TensorRT, Docker; NICE guidelines in FAISS, with Chroma used in testing), so audio stays on-device (NHS information governance).
- Cut Whisper word error rate (WER) by 20.7% and 24.4% on held-out real MDT recordings with silence-aware preprocessing, Bayesian hyperparameter search (Optuna) and noise-adaptive decoding; on augmented audio it came within 0.58 points of a commercial clinical ASR benchmark (Amazon Transcribe Medical).
- Designed an expert-labelled evaluation (66 interventions across 6 cases, two oncology experts; McNemar and paired bootstrap tests): the local MedGemma-RAG model recalled 2.3 times more MDT-concordant treatments than ChatGPT-5.2 (0.318 vs 0.136, $p = 0.020$), with half its hallucination rate under reasoning-first prompting.

COVID-19 household transmission estimated from social media, University College London 2021 – 2023

- Fine-tuned three COVID-Twitter-BERT (CT-BERT) classifiers with Hugging Face Transformers on 7,894 labelled tweets (10-fold cross-validation; AUC 0.907, 0.934, 0.768) to detect self and household cases.
- Built the pipeline from 1.23M candidate users to a geolocated UK cohort of 58,555 users and 107.9M tweets (Jan 2020 – Feb 2022); code public on GitHub (COVID_SAR).
- Estimated a household secondary attack rate of 0.16 (95% CI 0.14–0.18) and a mean serial interval of 6.49 days with a probabilistic model, and showed household infections were under-reported by about 44%.

Unsupervised misinformation detection, IIT Roorkee with Intel India 2020

- Designed an unsupervised framework (JoSE spherical embeddings, spherical k-means, user-credibility signals) to find misinformed content and users in 62M COVID-19 tweets; F1 0.899 (PolitiFact) and 0.965 (GossipCop), above all six supervised and clustering baselines (IEEE BigData 2020, oral).

Publications

Journal Articles

- **A. Dhiman**, E. Yom-Tov, L. Pellis, M. Edelstein, R. Pebody, A. Hayward, T. House, T. Finnie, D. Guzman, V. Lampos, Virus Watch Consortium, I. J. Cox. Estimating the household secondary attack rate and serial interval

of COVID-19 using social media. *npj Digital Medicine* 7(1):194, 2024. doi:10.1038/s41746-024-01160-2
- **A. Dhiman**, D. Toshniwal. AI-based Twitter framework for assessing the involvement of government schemes in electoral campaigns. *Expert Systems with Applications* 203:117338, 2022. doi:10.1016/j.eswa.2022.117338
- **A. Dhiman**, D. Toshniwal. A Twitter framework to assess the effectiveness of Indian government campaign. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, 2022. doi:10.1145/3490503
- **A. Dhiman**, D. Toshniwal. An approximate model for event detection from Twitter data. *IEEE Access* 8:122168–122184, 2020. doi:10.1109/ACCESS.2020.3007004

Conference and Workshop Papers

- **A. Dhiman**, D. Toshniwal. An unsupervised misinformation detection framework to analyze the users using COVID-19 Twitter data. *IEEE International Conference on Big Data (BigData)*, 2020 (oral). doi:10.1109/BigData50022.2020.9378250
- **A. Dhiman**, D. Toshniwal. An enhanced text classification to explore health based Indian government policy tweets. *DSHealth Workshop at ACM SIGKDD 2020* (poster). arXiv:2007.06511
- **A. Dhiman**, D. Toshniwal, S. Somani, P. Malik. Cleanliness campaign vs sanitation related diseases: Are they parallel in public perspective? *epiDAMIK Workshop at ACM SIGKDD 2018*, London (poster).
- **A. Dhiman**, D. Toshniwal. Do public and government think similar about Indian cleanliness campaign? *SIMBig 2018*, CCIS, Springer, pp. 367–380, 2019 (oral). doi:10.1007/978-3-030-11680-4_34

Preprints and Under Review

- **A. Dhiman**, F. Haque, I. Muazzam, K. Grover, L. B. Smith, W. S. Jones. Development and feasibility evaluation of an edge AI as medical device system for breast cancer multidisciplinary team meetings. arXiv preprint, 2026 (journal version submitted to PeerJ). doi:10.48550/arXiv.2608.22108
- **A. Dhiman**, M. Bano, L. Marino, T. Chen, M. Doyle. Machine learning based suicide-related risk assessment using structured NHS clinical data: A case control study. Under review at *Archives of Suicide Research*.

Skills

- **Languages:** Python. **ML and NLP:** PyTorch, TensorFlow, scikit-learn, XGBoost, SHAP, Hugging Face Transformers, transformer fine-tuning, text classification, clustering, embeddings (JoSE)
- **LLM and GenAI:** Large Language Models (MedGemma), Retrieval-Augmented Generation (RAG), FAISS, Chroma, prompt engineering, quantised inference (llama.cpp), speech recognition (Whisper), LLM evaluation
- **Data and Infrastructure:** NVIDIA Jetson, CUDA, TensorRT, Docker, Optuna, Google Colab, Git and GitHub
- **Methods and Domain:** experiment design, statistical testing (McNemar, bootstrap), cross-validation, probabilistic modelling, NHS clinical data, epidemiology, information governance

Education

PhD, Computer Science, Indian Institute of Technology (IIT) Roorkee, India — Mar 2017 – May 2022
Thesis: *AI based Analysis of Social Media Data for Public Health Domain*. Supervisor: Prof. Durga Toshniwal. Part-time from Jun 2021; viva May 2022.

MTech, Computer Science, National Institute of Technology (NIT) Kurukshetra, India — Aug 2014 – Jul 2016

BTech, Computer Science (Honours), Kurukshetra University, India — Aug 2010 – Jul 2014

Grants and Funding

- **Awarded and delivered:** Humber and North Yorkshire Cancer Alliance, Cancer Research and Innovation Award 2025–26: on-device AI for breast cancer MDT meetings. Co-Investigator, £41,707, Sep 2025 – Mar 2026.
- **In development:** Arts and Humanities Research Council (AHRC) bid on machine learning for 20th-century newspapers, with King's College London and Sheffield. Co-Investigator, about £500,000.
- **Awarded:** Intel conference-fee funding (2021); Microsoft Research and IIT Roorkee travel grants to present at ACM SIGKDD 2018, London (2018); full fee waiver, Lisbon Machine Learning School (2020).

Professional Service

- Peer reviewer for BMJ Health & Care Informatics, PLOS ONE, Data Mining and Knowledge Discovery, EPJ Data Science and Multimedia Systems; faculty research ethics reviewer (10+ applications).

Teaching and Supervision

- Module Leader (Applied AI; Big Data and Data Mining) across six core MSc AI modules (cohorts up to 121), incl. Python programming and NLP in healthcare; moved workshops from TensorFlow to PyTorch and Google Colab.
- Supervised about 40 MSc projects (LLMs, explainable AI), with supervisees in the cohort top ten in two intakes; lead the 7-member Digital Transformation Group on responsible generative AI.